Engineer's Logbook

for

eSTEM Challenge 2026

NJIT
The Center for
Pre-College Programs

My Name _____

Team Name _____

Welcome to the NJIT eSTEM Challenge

We are excited to have you participate in the 2026 eSTEM Challenge.
We created the challenge to:

- Get students excited about STEM – *Science, Technology, Engineering and Mathematics*
- Help students learn more about a specific STEM topic
- Guide students on how to be successful in a STEM challenge
- Create ways for students to have fun while they are learning

This Engineer's Logbook will help you understand how engineers work and think – the "E" in STEM. Logbooks are very important to engineers, as they serve as a place for engineers to document thoughts and ideas, work through problems, record data, and test different solutions. So, be sure to put your name on the logbook and complete each section as you move through the various steps.

The Challenge will help you learn about the other letters in STEM – Science, Technology and Math. Visit our website for a complete list of rules. Following the rules in a challenge is very important so everyone is treated fairly.

Rules, Updates, and Submissions

2026 eSTEM Resources

Part of following the rules is getting things done on time. See the website for dates for completing the parts of the challenge. Mark them on the calendar in this logbook. You will make a plan with your team for getting things done on time.

Dr. Devonii Reid

Director of Early College Preparatory Programs

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Table of Contents

Engineers keep a list of the things they put in their Logbooks. They give short titles to the things on a page to help them remember what is there. We've given you a start. When you finish a page, add an entry to the table of contents.

Page Number	Contents of the Page
2	Table of Contents
3	Starting Our Work - Working as a Team
4	What is the Challenge? Important Dates
5	How Do Engineers Work? Using the Logbook

Starting Our Work

Good Day, Engineers!

Engineers are people who think of ways to solve problems to make our lives better. They work on teams. Your new job is to work on a team to complete a design challenge. At the end of the challenge, your team will show what you've learned and how you have solved the challenge.

1. Begin by choosing a **team name**. Think of your ideas for a name. Write them down. Share with your teammates. Decide on the best name and write it on the third line.

   Names of my teammates:

   _____

   My ideas for a team name:

   _____

   Our team name is:

   _____

2. You're off to a great start! Now to review the ground rules. Work with your team to make a <u>short</u> list of rules that the team will follow. Here are some ideas, but you can make your own.
   a. Think before sharing with your team.
   b. Listen to everyone because everyone has good ideas.
   c. Ask for help when your team is stuck.
   d. Use materials with care & clean up when you're done.

   _____

   _____

   _____

   _____

It's time to take on your Challenge! Go to the next page in the Logbook.

The eSTEM Challenge

Now that you know *how* you will work, you need to know *what* you will do.

Your Team Challenge

Go to 2026 eSTEM Resources to find all the rules for the challenge.
Write in your own words what you have to do to successfully complete the challenge.

_____

_____

_____

_____

Write in the calendar the **due dates** for the parts of the challenge and when you will work.

2026 eSTEM Challenge Important Dates

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		January 20 **Challenge Kick-off**	21	22	23	24
25	26	27	28	29	30	31
February 1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
March 1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19 **eSTEM Challenge @ NJIT**		

How Do Engineers Work?

To be an engineer you will need to know how they work.

Engineers are people who think about, design, make, and test structures, machines, systems, and materials of all kinds. To do their work they must develop, test, install, configure and find the “bugs” in many things. It’s easier to fix something or make something better that you have created if you keep track of what you have thought about, tried out, and fixed before. Engineers do this in a special kind of Logbook, an **Engineer’s Logbook**.

All kinds of scientists, mathematicians, computer specialists, writers, musicians, and artists keep Logbooks. There can be words, sketches, and things pasted in the Logbook – anything that will help the person and other people understand and keep track of ideas.

You are going to keep an **Engineer’s Logbook** for this project. It will help you organize your ideas, and it will help others who look at your work.

What Is an **Engineer’s Logbook**?

An Engineer’s Logbook **IS**:

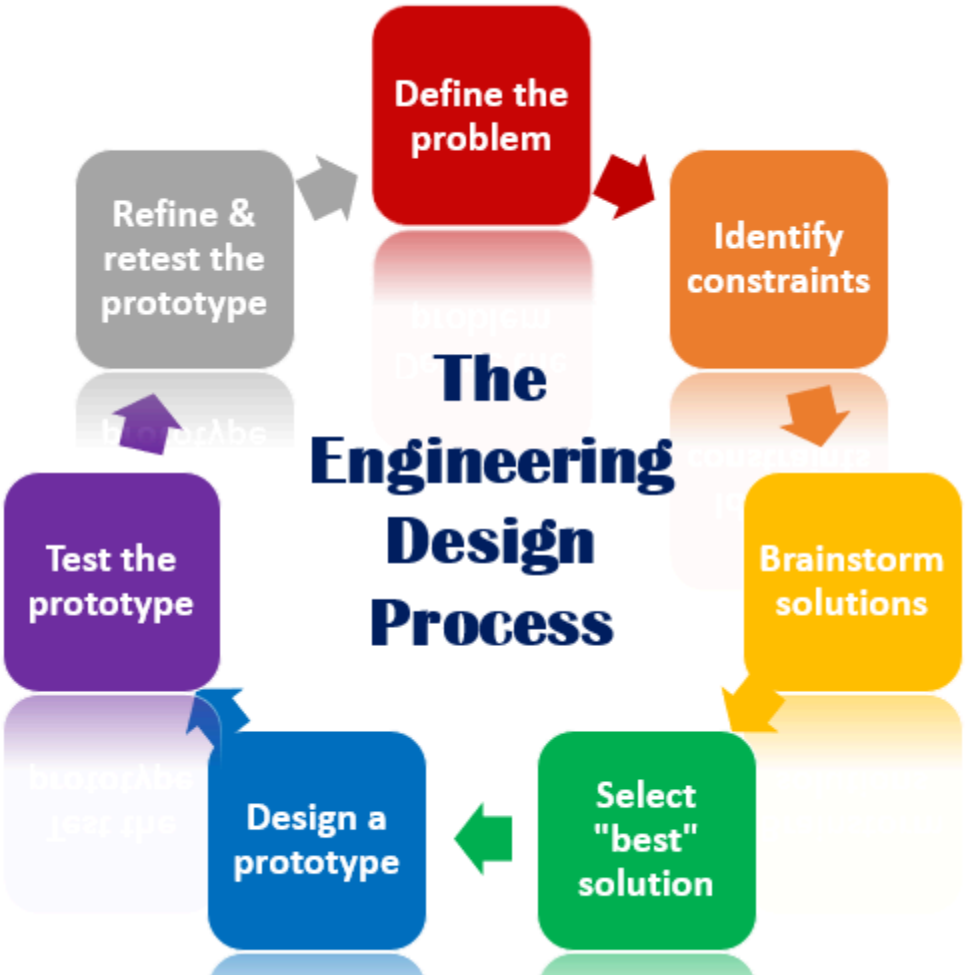
- A place to write down what you are thinking about a problem.
- A way to prove that something is your own idea.
- Always bound – pages can’t be taken out or added.
- Always labeled with the date – stuff is entered in date order.
- Always page-numbered with no skips.
- Never erased – we ALL make mistakes and change our minds; just ~~draw a line through~~, then write what you wanted to say or write “See page _” to send someone to a later page where you put a correction or new idea.
- Full of sketches you make and pictures you cut and paste or tape in from other places.
- Written so you and someone else can read it – even a year later!
- Full of thoughts about what you’ve done (*reflections*) and new ideas.

An Engineer’s Logbook **IS NOT**:

- Filled with anything other than the problems you are trying to solve in engineering.
- A place to write notes to friends or put stuff about other classes or doodles.

What is the Engineering Design Process?

Engineers take on many business and community challenges. In their attempts to resolve a challenge, engineers gather information, sketch out ideas, and test possible solutions. These action steps are part of the Engineering Design Process (EDP).



To begin, start at “Define the Problem” and move around the circle to the right, like the hands on a clock. But look at the arrows! Sometimes you will want to go forwards, other times you will want to move backwards and repeat a step - maybe more than once. Engineers do that all the time! The back and forth is necessary to modify and refine ideas.

In this Logbook, you will work on each of the seven steps. Here's what they mean:

Define the problem: Take some time to ask questions. What is the problem? How do you know it's a problem? How would you describe the problem? What previous work has been done to resolve the problem?

Identify constraints: Conduct your research. What are the limitations on any proposed solution? Who is setting these constraints? How will the constraints restrict your proposed solution?

Brainstorm solutions: How could you solve this problem? What are some other ways to solve the problem? What are your teammates' ideas? Gather as many ideas as possible.

Select the "best" solution: Evaluate all ideas. Select a favorite. As a team, review all the ideas the team proposed. Discuss the benefits and drawbacks of each possible solution. Come to an agreement on the best solution to pursue considering the earlier identified constraints.

Design a prototype: Determine what needs to be done to convert the proposed solution from an idea into a working prototype. Make a list of the steps you will follow. Draw a sketch – or more than one. Label the sketches. Get all the things you need to build the prototype.

Test the prototype: Once you have built your prototype, test it to see if it works as you wanted. If it doesn't, reflect on what might have happened. Does the current design need revision? If so, in what ways? Do you need to select a new design from your other proposed ideas?

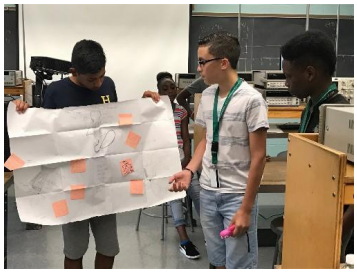
Refine & retest the prototype: This is the opportunity to make the prototype. What worked in your solution? What didn't work and how can you fix it? How could you make it better – easier, smoother, faster, cheaper? Do you need to change the steps? Do you need to change the sketch(es)? Test it again.

You will need to think about these steps over and over as you work on the Challenge. Go to the inside of the back cover to remind yourself of the seven steps.

Remember to fill in the Table of Contents!

Reflect on Your Work

Date: _____



Another thing that Engineers do is think about what they have done well and what they can do better. That's called ***reflecting***.

Complete this page with your team. There are other pages like this in your Logbook. Look back at all the pages you have finished before you start your next week.

1. What work did we do today?

   _____

   _____

2. What did ***I*** do well?

   _____

   _____

3. What did ***we*** do well as a team?

   _____

   _____

Make a plan for next time:

4. Next time, we will work even better as a team by doing:

   _____

   _____

5. What work do we want to do next time?

   _____

   _____

6. What stuff will we need?

   _____

   _____

Remember to fill in the Table of Contents!

Define the 'Problem'

Date: _____

Ask Questions: What is a healthcare or public health preparedness, response, or recovery challenge you see in your community?

_____

_____

_____

_____

What have you seen or heard that makes you believe this is a challenge?

_____

_____

_____

_____

How can you learn more about this challenge? To whom could you talk to learn more?

_____

_____

_____

_____

How do you think you can help address the challenge?

_____

_____

_____

_____

Remember to fill in the Table of Contents!

Identify the Constraints

Date: _____

There are always some limits (*constraints*) on what you can do or use to solve your challenge or problem. They may not say, "*This is a limit.*" Sometimes they are hidden. Look for words about size, temperature, time, etc.

What are the limits (*constraints*) for this Challenge? Write them in the Brainstorm first. Put numbers beside them to show the order with the most important ones first. Then make an Organized List.

Brainstorm all possible project constraints.

_____

_____

_____

_____

_____

_____

Identify which constraints may be the biggest obstacles to your project.
(List the constraints in order from LARGEST constraint to SMALLEST).

_____

_____

_____

_____

_____

_____

Next, you will need to learn more about how you want to enhance healthcare and public health preparedness, response, or recovery.

Talk to ‘Stakeholders’

Date: _____

Find out all you can before you make your plan. Learn what others know about your problem. Think about how others are affected by your identified challenge.

Write down what you learn. You may also sketch or draw what you learn on the right side.

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Conduct Library Research

Date: _____

To confirm that an issue is truly a challenge for a community, engineers must conduct research. This research must include information from credible and reliable sources.

A great place to find credible and reliable sources is the library.

First, discover what resources are available for your research.
(*Books? Magazines? Journals? Databases?*)

Write down the specific name and location of these resources.

_____

_____

_____

_____

_____

_____

Next, using these resources, find specific pieces of information that describe or support the healthcare or public health preparedness, response, or recovery challenge you identified in your community. ***Provide formal citations for EACH piece of information***.

Citation format:
Author last name, Author first name. (Year published). Title of book. Name of publisher.

Citation example:
Showers, Paul. (2000). *Hear your Heart.* Harper Collins Publishing.

_____

_____

_____

_____

_____

Additional resources can be listed on the following page.

Library Research (continued)

Additional citations for your information can be listed here.

Reflect on Your Work

Date: _____

Another thing that Engineers do is think about what they have done well and what they can do better. That's called ***reflecting***.

Complete this page with your team. There are other pages like this in your Logbook. Look back at all the pages you have finished before you start your next week.

1. What work did we do today?

   _____

   _____

2. What did ***I*** do well?

   _____

   _____

3. What did ***we*** do well as a team?

   _____

   _____

Make a plan for next time:

4. Next time, we will work even better as a team by doing:

   _____

   _____

5. What work do we want to do next time?

   _____

   _____

6. What stuff will we need?

   _____

   _____

Remember to fill in the Table of Contents!

Sometimes engineers only use one side of the page in their logbooks. They save the other side for extra notes and drawings. If they do not use the page at all, they may draw a line from top to bottom and write their name on the line to show they meant to leave it blank. Or they might write:

This page is blank.

Brainstorm Solutions

Date: _____

Now that you and your team have identified a healthcare or public health preparedness, response, or recovery you want to enhance for the eSTEM Challenge, you are ready to begin brainstorming possible improvements!

Ask yourself, "*What is one way I could improve this issue?*" "*What are other ways that the issue might be resolved?*"

Write or draw all possible solutions in the space below.

_____

_____

_____

_____

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_____

Compare Solutions

Date: _____

Talk to your teammates. What possible enhancements did they develop? What led them to think of these solutions?

Choose three or four of your favorite ideas from your teammates. Write or draw their ideas in the space below.

_____

_____

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Choose a Solution

Consider your ideas and your teammates' ideas. Together, as a team, choose one idea to build and develop as an enhancement to the healthcare or public health preparedness, response, or recovery effort you identified.

Describe in detail the idea your team will build and develop.

_____

_____

_____

_____

_____

_____

_____

Draw and ***label*** a sketch that shows what your team will try to make. Write the size (*dimensions*) of each part. Write one or two words to show what each part is made from. If there is anything in your drawing that isn't part of what you will make, like water or the sun, label them, too. You can draw arrows from the words to the drawing.

STOP! **Look back at page 10. Compare your plan and the limits (*constraints*). Are they a good match? If not, what do you need to fix?**

More About Your Solution

What do you need to make your plan work? List everything you can think of!

People: Who are the members of your team? List each teammate's name and job(s). Do you need anyone in addition to your teammates? What will that person do?

_____

_____

_____

_____

_____

Materials (stuff): What type of materials do you need? Where will you get the materials you need to build your prototype?

_____

_____

_____

_____

_____

Technology: What technology will you need to collect your data and build your prototype? How will you access the technology you need?

_____

_____

_____

_____

_____

Budget for Your Solution

Budget: Make a list of the estimated costs (*budget*) of the “found items” you used to design your prototype (add rows to the table if you need more):

Item	Unit Cost	Number of Units	Total Cost

List the items your team purchased with the Amazon gift card provided by NJIT. Include here a copy of the receipt showing how much each item cost to purchase.

Item	Unit Cost	Number of Units	Total Cost

Reflect on Your Work

Date: _____

Continue ***reflecting*** on your work as an individual and a team. What has gone well? What would you like to improve upon?

Look back at all the pages you've finished before you start your next week.

1. What work did we do today?

   _____

   _____

2. What did ***I*** do well?

   _____

   _____

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   _____

   _____

Make a plan for next time:

4. Next time, we will work even better as a team by doing:

   _____

   _____

5. What work do we want to do next time?

   _____

   _____

6. What stuff will we need?

   _____

   _____

Remember to fill in the Table of Contents!

This page is blank - or you can use it for more notes or sketches.

Collect Your Data

Date: _____

Use this space to record the data collected by the Micro:bit.

You can draw a table or chart or cut and paste a table or pie chart you created electronically.
(*pie charts, bar graphs, line charts, etc.*)

Since collecting data is part of the Engineering Design Process AND a requirement of the eSTEM Challenge, you MUST include the data you collected on this page.

This page is blank - or you can use it for more notes or sketches.

Analyze Your Data

Date: _____

Use this space to show how you analyzed the data collected by the Micro:bit. Provide summary statistics about your data. Then describe what the data is telling you and say how the data will drive the changes made to your prototype.

You may *draw* a table or chart of your data. You may also *cut and paste* a table or pie chart you created electronically.

Reflect on Your Work

Date: _____

Continue ***reflecting*** on your work as an individual and a team. What has gone well? What would you like to improve upon?

Look back at all the pages you've finished before you start your next week.

1. What work did we do today?

   _____

   _____

2. What did ***I*** do well?

   _____

   _____

3. What did ***we*** do well as a team?

   _____

   _____

Make a plan for next time:

4. Next time, we will work even better as a team by doing:

   _____

   _____

5. What work do we want to do next time?

   _____

   _____

6. What stuff will we need?

   _____

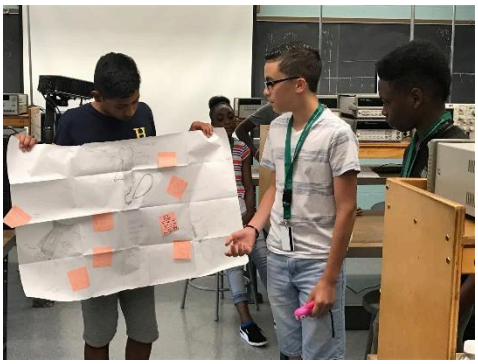
   _____

Remember to fill in the Table of Contents!

Revise Your Plan

Date: _____

Most of the time, engineers' first tries don't work! You may need to "fix" your first idea or make it work better. Talk to your team:



- What worked?
- What didn't work?
- Do you need to change your idea?
- Are the steps OK or do you need to fix them?
- Do you need to change the sketch(es)?
- If it worked, can you make it better?
- How much does your model cost? Could you make it cost less and work just as well?

Make notes and sketches in the space below.

This page is blank - or you can use it for more notes or sketches.

Retest your Prototype and Collect New Data

Date: _____

Use this space to record the data collected by your prototype. You may *draw* a table or chart of your new data. You may also cut and paste a table or chart you created electronically.

Reflect on the Overall Project

Answer these questions about the overall design process.

About the Challenge

1. What did we do well to respond/solve the challenge?

   _____

   _____

2. What would we change if we did it again?

   _____

   _____

About our Team

3. How did we work as a team? What did we do well together?

   _____

   _____

About Myself

4. What did I learn?

   _____

   _____

5. What do I want to learn about next?

   _____

   _____

6. What can I use from this experience to help me learn it?

   _____

   _____

Prepare to Communicate Your Results

Teams are encouraged to complete this template as a way to prepare for the visual display and presentation portion of the challenge.

Design Summary Outline

Introduction: Introduce your team and your prototype.
Problem Statement: Briefly describe the healthcare problem facing your community and why it was chosen for the Challenge.
Topic Research: Discuss key information about the topic that has been gathered through multiple sources of information.
Project Goals: List project goals and show how they are linked to and will adequately meet the user's needs.

Key Features of Design: List key features of the design and show how they adequately meet project goals.
Prototype Testing: Explain the details of how, where, and when the prototype was tested.
Revised Features of Design: List key features of the design that were revised to more adequately meet project goals.
Status of Project: Describe the current status of the project and discuss potential next steps.
Reflection: Discuss personal growth and insights about designing for others and helping them overcome challenges. Also, include discussion of any increased understanding of the Engineering Design Process.

STOP! **Make a copy of this outline to bring with you the day of the Challenge!**
The outline will be one of the resources students can use when presenting to judges.

Resources We Found and Want to Remember

Write the URLs or names of resources that helped you.

Notes

This is your last entry in the Table of Contents!

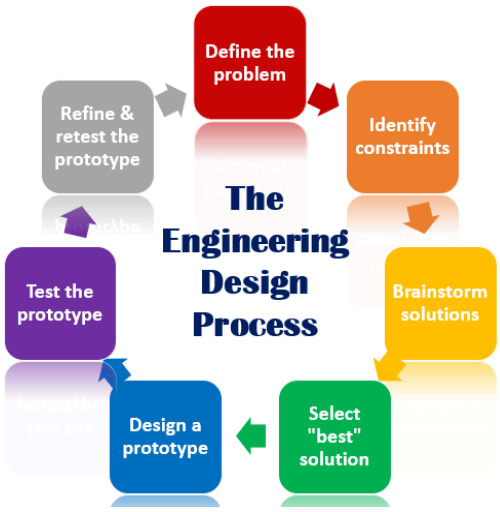
This page is blank - or you can use it for more notes or sketches.

This page is blank - or you can use it for more notes or sketches.

This page is blank - or you can use it for more notes or sketches.

How Do Engineers Solve Problems?

Engineers make a plan to solve each problem. Here is a plan engineers might use. Start at the top and move around the circle like hands on a clock move. But look at the arrows! Sometimes you will want to go back and forth – maybe more than once. Engineers do that all the time!



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NJIT The Center for Pre-College Programs

eSTEM

To Contact Us:

Early College Preparatory Programs

4th Floor Campbell Hall

University Heights Newark, NJ 07102-1982

973-596-3550

Dr. Devonii Reid

Director of Early College Preparatory Programs

Mellina Truncale

Assistant Direct of Curriculum & Instruction

eSTEM Coordinator